Distinct 1+1 branch: validity preservation

Private field-level proof; independent review pending

31 October 2026

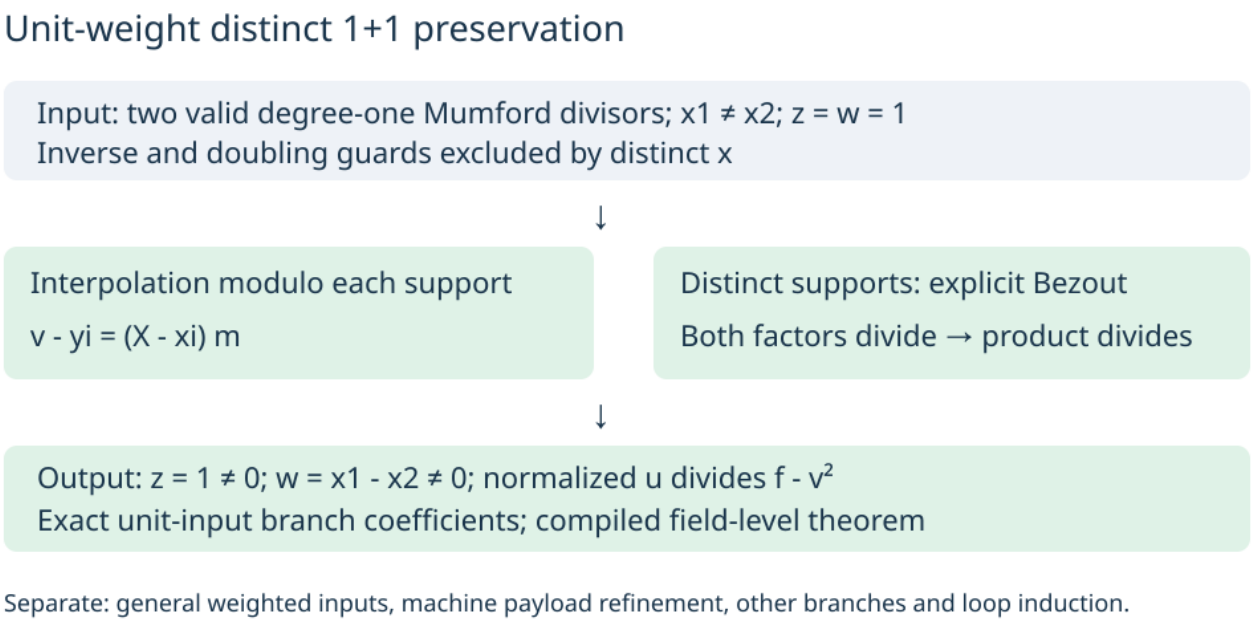
The unit-weight distinct branch of the actual add11 formula preserves normalized Mumford validity and nonzero output weights. The proof assumes two valid degree-one inputs and distinct x-coordinates. A companion theorem checks branch precedence: inverse and doubling both require equal x.

Exact formula

For inputs $(x_1, y_1), (x_2, y_2)$ with unit weights, the source returns

$$q = -(x_1 + x_2), \quad r = x_1 x_2, \quad s = y_1 - y_2, \quad t = x_1 y_2 - x_2 y_1, \quad z = 1, \quad w = x_1 - x_2.$$

The normalized support is $(X - x_1)(X - x_2)$ and the ordinate is the interpolation line $v = (s/w)X + t/w$. The proof derives each linear support's divisibility of $f - v^2$, then uses an explicit constant Bezout identity to obtain product divisibility.



Separate: general weighted inputs, machine payload refinement, other branches and loop induction.

Checks and controls

Two proof stages compile. Separate readback and controls are retained. The positive control uses $f = X^5 - X + 1$ over $\mathbb{Q}$ with $(0,1),(1,1)$, explicitly proves both degree-one input divisibilities and applies the full normalized output theorem. Distinct and doubling selection and overlapping inverse precedence are checked. Equal x yields zero output w, marking the required guard boundary.

The initial bounded API probe failed because polynomial evaluation/factor-theorem modules were absent from the pinned minimal library. The preserved amendment uses degree-one Mumford divisibility directly and proves the Bezout step without new dependencies. Successful proof logs retain stylistic linter warnings. The first control compile needed a noncomputable polynomial abbreviation; the corrected control passes and the failed attempt remains preserved. The first PDF render lacked the amssymb import; its logs are also retained.

Source and scope

Exact source: projective_law_gate.rs add11 lines 511–532, from_affine lines 569–580 and dispatcher lines 606–625; projective_state_collector.rs lines 153–160. Retained hashes 5976e7f3 and 96246448 match current HEAD in the preceding constructor packet. Degree-one from_affine assigns unit weights, which is the input specialization proved here.

Input validity means $(X - Cx_i)$ divides $f - (Cy_i)^2$. The theorem covers any field and polynomial with those hypotheses. Smooth quintic/odd-characteristic assumptions belong to the intended Jacobian interpretation. No native j.valid refinement, UInt64/counter payload proof, generic weighted 1+1 law, doubling law, divisor-class equality or loop induction is claimed. No scientific trial or cost ratio was produced.

Evidence and prior work

Add11.lean, Controls.lean, Readback.lean, raw compile receipts, REPORT.md, status.json and the source manifest accompany this PDF. Prior InvariantBranches does not contain this output congruence. Literature: Ozbay Gurler–Hisil, DOI 10.1007/s13389-024-00351-7, sections 2–4. The result is a local implementation-preservation connection.